# AI-Cost Factor Model — Counterfactual Risk Quantification (Subscription-Aware Pilot, v0.2.10)

Demonstration-Grade Pilot Iteration Closure

Abrigo Analytics — Iteration `dev_ai_cost_v2`

2026-05-18

**Abstract**

We report the closure of the `dev_ai_cost_v2` iteration under spec v0.2.10 (post audit-econ Delphi). On a single-user Claude Code JSONL panel covering 2026-01-06 to 2026-05-14 ($N = 28$ weekday rows after first-differencing), the FX-vol contribution to AI-cost variance is empirically approximately zero: under the conservative covariance-attribution rule of Politis and Romano (1994), the FX share of $\mathrm{Var}(\Delta \ln \mathrm{NotionalCost}^{COP})$ point-estimates to 0.0000277 with a 90% stationary-bootstrap confidence interval of $[-3.4 \times 10^{-6}, +4.1 \times 10^{-5}]$. The pre-registered behavioral subscription-inelasticity test (R4-S3-USD), an HAC-OLS regression of $|\Delta \ln \mathrm{NotionalCost}^{USD}|$ on lagged $|\Delta \ln \mathrm{USDCOP}|$ and $|\Delta \ln \mathrm{Tokens}|$ with Newey and West (1987) bandwidth $L = \lfloor T^{1/3} \rfloor = 3$, is *POWER-HALTED* under the canonical lagged-tokens partialling recipe: measured power at MDES = 0.40 residual-SD is 0.1745, below the *demonstration-grade* floor of 0.50 ($N_{\min} = 38$, POWER_MIN = 0.50 — below project defaults $N = 75$ / 0.80). The iteration is paused pending an $N \geq 75$ panel and makes *no* population-level transmission claim. We recommend the framework owner pivot to (i) data accumulation, (ii) the R6 continuous-stream sibling iteration, or (iii) a target population where the FX channel is ex-ante stronger.

## 1 Introduction

The Abrigo project's operating framework studies permissionless on-chain convex hedges aimed at the Latin American wage-earner-to-capital transition. Each iteration is a $(Y, M, X)$ triple in which $Y$ is an outcome surfacing the target population's exposure, $M$ is a Panoptic-eligible position settling the empirical risk, and $X$ is the major risk currently blocking the transition. The present iteration is a *candidate-X identification step*: does FX volatility (USDCOP) materially drive AI-cost burden risk for LATAM developers? The R5 PRIMARY arm answers a descriptive risk-decomposition question (FX share of cost-burden variance); the R4-S3-USD AUXILIARY arm answers a behavioral question (does USD-side cost vol respond to FX vol despite zero marginal cost under a subscription regime).

**Subscription-aware proxy.** The pilot subject (the repo maintainer) is on a Claude Pro/Max subscription. Their realized USD cost is approximately flat per month and FX exposure collapses to a single billing-day point. Their *notional* cost path — rate-card price × token usage per the LiteLLM SHA-pinned rate table (LiteLLM, 2025) — serves as a fair proxy for what a non-subscribed pay-per-API LATAM developer would have paid. This is a strong assumption (equivalent demand under both billing regimes) and is documented as a Section 6 threat.

**v0.2.10 audit-closure context.** The v0.2.9 verdict (PARTIAL-FAIL_TO_REJECT under a contemporaneous power-recipe pin) was withdrawn following an audit-econ Delphi (2026-05-18). Four critical / high-severity amendments enter v0.2.10: (#4) Z-arms relabeled as *same-ratio* corroborations of R5 rather than independent measurements; (#5) demonstration-grade vs. verdict-grade scope split with mandatory headline disclosure; (#6/#7) R4-S3-COP's REGIME-CONDITIONAL FAIL removed from the corroboration list as uninformative; (#8) the canonical power-calculation recipe reverted to lagged-tokens partialling, matching the regression's own $k = 1$ lag structure. Under the lagged recipe, the power gate fires and routes R4-S3-USD to POWER-HALT.

# 2 Data

The panel is constructed from local Claude Code JSONL session logs on the pilot subject's machine. Costs are computed as rate-card × token-counter usage across the five preserved token categories (input, output, cache-creation-ephemeral-5m, cache-creation-ephemeral-1h, cache-read) using the LiteLLM rate table pinned by both commit SHA and file SHA-256 (dual-pin enforcement, audit-econ closure #3). USDCOP is the Banrep daily TRM. The build window observed is 2026-01-06 to 2026-05-14.

**Counters and parity.** Per audit-econ closures #1, #9, and #10, the row-drop bookkeeping is split into named counters (Codex-commingling filter at parse, non-Anthropic-model upstream filter, holiday-Monday weekday-mask surfacing). ccusage cost-parity on the build window is 0.9994× on raw cost totals; the combined-window aggregate ratio of 0.5386× reflects the audit-disclosed treatment of historical Codex commingling outside the build window and is documented separately in `DATA_PROVENANCE.md`.

**Demonstration-grade pin.** The project-level anti-fishing invariants pin $N_{\min} = 75$, POWER_MIN = 0.80, MDES_SD = 0.40. v0.2.0 CORRECTIONS-G + CORRECTIONS-J relaxed the first two to 38 and 0.50 without a Pair-D–style preserved-power proof. v0.2.10 Amendment #5 formalizes this as a separate *demonstration-grade* surface: relaxed floors are permitted, but every consumer (verdict memo, this write-up, downstream M-design hand-off) must disclose the relaxation in the headline. The present iteration is demonstration-grade with $N = 28$ observed (sub-floor relative to the demonstration-grade $N_{\min} = 38$). No verdict-grade claim is made.

# 3 Methodology

## 3.1 R5 PRIMARY: descriptive counterfactual risk quantification

The realized COP-denominated cost-burden return is the log first-difference of $\text{NotionalCost}_t^{COP} = \text{NotionalCost}_t^{USD} \cdot \text{USDCOP}_t$. Under Politis and Romano (1994) stationary bootstrap with expected block length $\lceil T^{1/3} \rceil$ and $B = 10{,}000$ replicates, we report the variance decomposition with *conservative covariance attribution* (CORRECTIONS-Q):

$$\text{Var}(\Delta \ln \text{NotionalCost}^{COP}) \;=\; \text{Var}(\Delta \ln \text{NotionalCost}^{USD}) + \text{Var}(\Delta \ln \text{USDCOP}) + 2\ \text{Cov}(\Delta \ln \text{NotionalCost}^{USD} \tag{1}$$

The headline FX share is $\text{Var}(\Delta \ln \text{USDCOP}) / \text{Var}(\Delta \ln \text{NotionalCost}^{COP})$ *excluding* the covariance term, which is reported separately as a diagnostic with its own bootstrap CI. The covariance-exclusion rule is conservative for the M-design gate: positive covariance under alternative attribution schemes would inflate the apparent FX share.

**R5 verdict logic.** None. R5 is descriptive — it produces a number with bootstrap CI consumed by the M-design step. The only pre-pinned quality threshold is the CI half-width on the FX share $\leq 0.15$, an M-design usability gate (CORRECTIONS-T), not a verdict gate.

### 3.2 R4-S3-USD AUXILIARY: behavioral subscription-inelasticity

The R4-S3-USD specification is

$$|\Delta \ln \mathrm{NotionalCost}_t^{USD}| = \alpha_0^{USD} + \alpha_1^{USD} |\Delta \ln \mathrm{USDCOP}_{t-k}| + \alpha_2^{USD} |\Delta \ln \mathrm{Tokens}_{t-k}| + u_t^{USD}, \tag{2}$$

with $k = 1$ primary and $k = 5$ sensitivity. The pre-pinned null is $\alpha_1^{USD} = 0$ under a *two-sided* test (CORRECTIONS-S): under the subscription regime, marginal USD cost is zero, so the behavioral channel should produce no response in either direction; rejection either way is the informative finding. HAC standard errors follow Newey and West (1987) with the Andrews (1991, Andrews)-style data-dependent bandwidth $L = \lfloor T_{\mathrm{lag}}^{1/3} \rfloor$, automated per Newey and West (1994).

**Power-HALT checkpoint.** Power is Monte-Carlo'd at MDES = 0.40 residual-SD on the post-control residual variance of $|\Delta \ln \mathrm{NotionalCost}^{USD}|$ after partialling $|\Delta \ln \mathrm{Tokens}|$. Per CORRECTIONS-U the power-HALT was expected to fire at $T = 38$. The canonical partialling-regressor recipe is *lagged tokens* (matching the regression's own $k = 1$ lag structure), per v0.2.10 Amendment #8; the contemporaneous-tokens recipe is reported for transparency but does not gate the verdict.

### 3.3 Z-arms sensitivity (diagnostic only)

Z-1 aggregates the same variance ratio at weekly (Z-1a) and monthly (Z-1b) horizons. Z-2-main back-casts the observed $\Delta \ln \mathrm{NotionalCost}^{USD}$ distribution onto the 2024–2025 daily TRM via stationary bootstrap ($B_{\mathrm{paths}} = 1{,}000$); Z-2-null repeats the loop with the 2026-Q1-Q2 TRM only for null calibration; Z-2-W conditionally activates a Winsorized robustness sub-arm if observed excess kurtosis exceeds 3.0. Z-3 is the escalation gate (OR over relative-5×-baseline and absolute-0.05 thresholds, with null-calibration suppression). **Per v0.2.10 Amendment #4, all Z-arms decompose the same algebraic variance ratio across different horizons / TRM windows and are *same-ratio corroborations* of R5, not independent measurements.**

## 4 Results

### 4.1 R5 PRIMARY

The headline FX share over the daily 2026-Q1-Q2 window ($T = 29$ daily rows, $N = 28$ post-first-diff) is reported in Table 1. The FX share's stationary-bootstrap CI half-width is $2.2 \times 10^{-5}$, three orders of magnitude inside the 0.15 M-design usability threshold. The covariance term is small, negative ($-0.002924$), and its CI crosses zero — consistent with a weak natural-hedge effect within the developer's behavior, but not significant.

### 4.2 R4-S3-USD: POWER-HALT

The R4-S3-USD primary regression at $k = 1$ yields $\hat{\alpha}_1^{USD} = -18.116$, HAC SE = 40.221, $t = -0.450$, two-sided $p = 0.6524$, with HAC bandwidth $L = \lfloor 28^{1/3} \rfloor = 3$. **These coefficients are reported for the record only; they do not adjudicate the null.** Under the canonical lagged-tokens partialling recipe, the measured power at MDES = 0.40 residual-SD is 0.1745,

Table 1: R5 PRIMARY variance decomposition — conservative attribution (CORRECTIONS-Q). 90% stationary-bootstrap CI, Politis and Romano (1994), $B = 10{,}000$, expected block length $\lceil T^{1/3} \rceil = 4$.

| Component | Point estimate | 90% CI lower | 90% CI upper |
|---|---|---|---|
| FX share (excl. cov) | 0.0000277 | $-3.4 \times 10^{-6}$ | $+4.1 \times 10^{-5}$ |
| Usage share (excl. cov) | 1.001078 | 0.998897 | 1.003730 |
| 2 × Cov (diagnostic) | −0.002924 | −0.008650 | +0.002815 |
| $\hat{\sigma}_{\Delta \ln \text{NotionalCost}^{COP}}$ | 1.659 | 1.216 | 2.125 |

below the demonstration-grade floor of 0.50. The contemporaneous-tokens recipe (diagnostic only) yields 0.7115. The verdict label is **POWER-HALT**; PARTIAL-REJECT and PARTIAL-FAIL_TO_REJECT are *not applicable* because the power gate fires before the $N$-floor gate. Table 2 summarizes.

Table 2: R4-S3-USD HAC-OLS results ($k = 1$ primary, $k = 5$ sensitivity) and power gate. Reported coefficients are for-the-record only; the verdict is POWER-HALT.

| | $k = 1$ (primary) | $k = 5$ (sensitivity) |
|---|---|---|
| $\hat{\alpha}_1^{USD}$ | −18.116 | −35.750 |
| HAC SE | 40.221 | — |
| $t$-statistic | −0.450 | — |
| Two-sided $p$ | 0.6524 | — |
| HAC bandwidth $L$ | 3 | 2 |
| Residual SD (lagged partialling) | 1.156 | |
| MDES effect $0.40 \times \sigma_e$ | 0.462 | |
| Power (lagged, **canonical**) | **0.1745** (gate: **FAIL**) | |
| Power (contemporaneous, diagnostic) | 0.7115 (gate: pass) | |
| **Verdict** | **POWER-HALT** | |

### 4.3 Z-arms (diagnostic, same-ratio)

Table 3 reports the same-ratio Z-arms. The Z-3 verdict is `NOT_ESCALATE`: no Z-arm trips the OR-rule ($\geq 5\times$ baseline or $\geq 0.05$ absolute). Per v0.2.10 Amendment #4, these arms are *not* independent measurements and are not counted as corroborating evidence for the headline R5 finding.

## 5 Discussion

**Regime-conditional reading.** For this user × this 2026-Q1-Q2 window, individual cost-burden volatility is dominated by token-burst usage variance, not by FX moves. The usage-share point estimate of 1.001 (with its CI tightly around 1.0) and the FX-share point of $2.77{\times}10^{-5}$ (CI containing zero on the lower end) together imply that the cost panel's variance, on this window, is algebraically explained by USD-side variance with a small negative covariance adjustment. The reading is descriptive, regime-conditional, and bounded to the n=1 proxy subject.

Table 3: Z-arms (same-ratio corroborations of R5; diagnostic-only).

| Arm | FX share | Notes |
|---|---|---|
| Daily baseline (R5) | 0.0000277 | primary |
| Z-1a weekly | 0.000073 | $T_w \approx 9$ |
| Z-1b monthly | 0.000099 | $T_m = 4$, TAGGED uninformative |
| Z-2-main backcast median | near baseline | 2024–2025 TRM, $B_{\text{paths}} = 1{,}000$ |
| Z-2-W (Winsorized) | conditional | activates iff excess kurtosis $> 3.0$ |
| Z-3 verdict | | `NOT_ESCALATE` |

**What R5 does and does not say.** R5 is Role A (individual-sizing) per CORRECTIONS-R. The implication for the M-design step is that an individual hedge for this user, at this notional burn rate, should be dominated by token-volume risk exposure rather than FX exposure. *This is not a population-level channel-existence claim.* Channel existence under the framework requires the R4-S3-USD behavioral test, which here POWER-HALTs.

**Why R4-S3-USD cannot be adjudicated.** The behavioral channel asks whether a subscription user nevertheless shifts token consumption in response to FX volatility (e.g., defensively reducing usage when COP weakens, despite the flat USD invoice). At $N = 28$ and HAC-corrected SE of 40.2 on $\hat{\alpha}_1^{USD}$, the regression cannot distinguish meaningful behavioral response from noise. The high $p = 0.6524$ is *uninformative* once the power gate has failed: with $B = 2{,}000$ simulated draws at MDES $0.40\sigma_e$, only 17% of draws reject under the lagged canonical recipe. The honest reading is "insufficient data," not "no effect."

**Why Z-arms cannot rescue R4-S3-USD.** Z-arms re-decompose the same R5 ratio across alternate time horizons and TRM windows. They corroborate the descriptive small-FX-share finding (regime-conditional) but carry no independent inferential weight on the behavioral channel. Counting them as independent evidence is fishing-by-redundancy under §7's anti-fishing invariants.

## 6 Limitations

1. **Sub-floor $N$.** Observed $N = 28$ is below the demonstration-grade $N_{\min} = 38$ (itself below the project default of 75). All findings are pilot-grade.
2. **Single-user pilot.** The panel is a single-user JSONL trace. No population-level inference is admissible.
3. **Subscription-as-proxy.** The notional-cost path substitutes for what a non-subscribed pay-per-API LATAM developer would have paid for equivalent usage. Equal demand under both billing regimes is a strong assumption.
4. **v0.2.10 audit-econ corrections context.** The withdrawn v0.2.9 verdict (PARTIAL-FAIL_TO_REJECT under the contemporaneous power-recipe pin) remains in git history for audit trail. The current write-up reports the lagged canonical recipe per Amendment #8.
5. **Regime-conditional reading.** The small-FX-share finding is conditional on the 2026-Q1-Q2 USDCOP regime and on the pilot subject's usage pattern. Generalization is not claimed.
6. **Same-ratio corroboration discipline.** Z-arms are not independent inferential evidence; they are reported as diagnostic-only.

## 7 Conclusion

The `dev_ai_cost_v2` iteration is paused at **demonstration-grade pending an $N \geq 75$ panel** — explicitly not closed-PASS and explicitly not closed-FAIL. The R5 descriptive finding (FX share $\approx$ 0%, regime-conditional) informs M-design priors under Role A but is not a framework-level channel-existence statement. The R4-S3-USD behavioral test POWER-HALTs; it cannot be adjudicated at current sample size and power.

Three pre-enumerated pivots are available to the framework owner: (i) continue accumulating data on the pilot subject ($\sim$ 12–18 months to reach $N \geq 75$ at observed cadence; regime-change risk material); (ii) pivot within the same $Y$ to the R6 sibling iteration on continuous-stream simulation, parked at v0.1.3 and ready for dispatch; or (iii) pivot to a different $Y$ (target population) where the FX channel is ex-ante stronger. The choice is the framework owner's. None of the three implies that the FX-channel hypothesis has been disconfirmed; only that this particular pilot, on this subscription user and this window, cannot adjudicate it.

## References


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